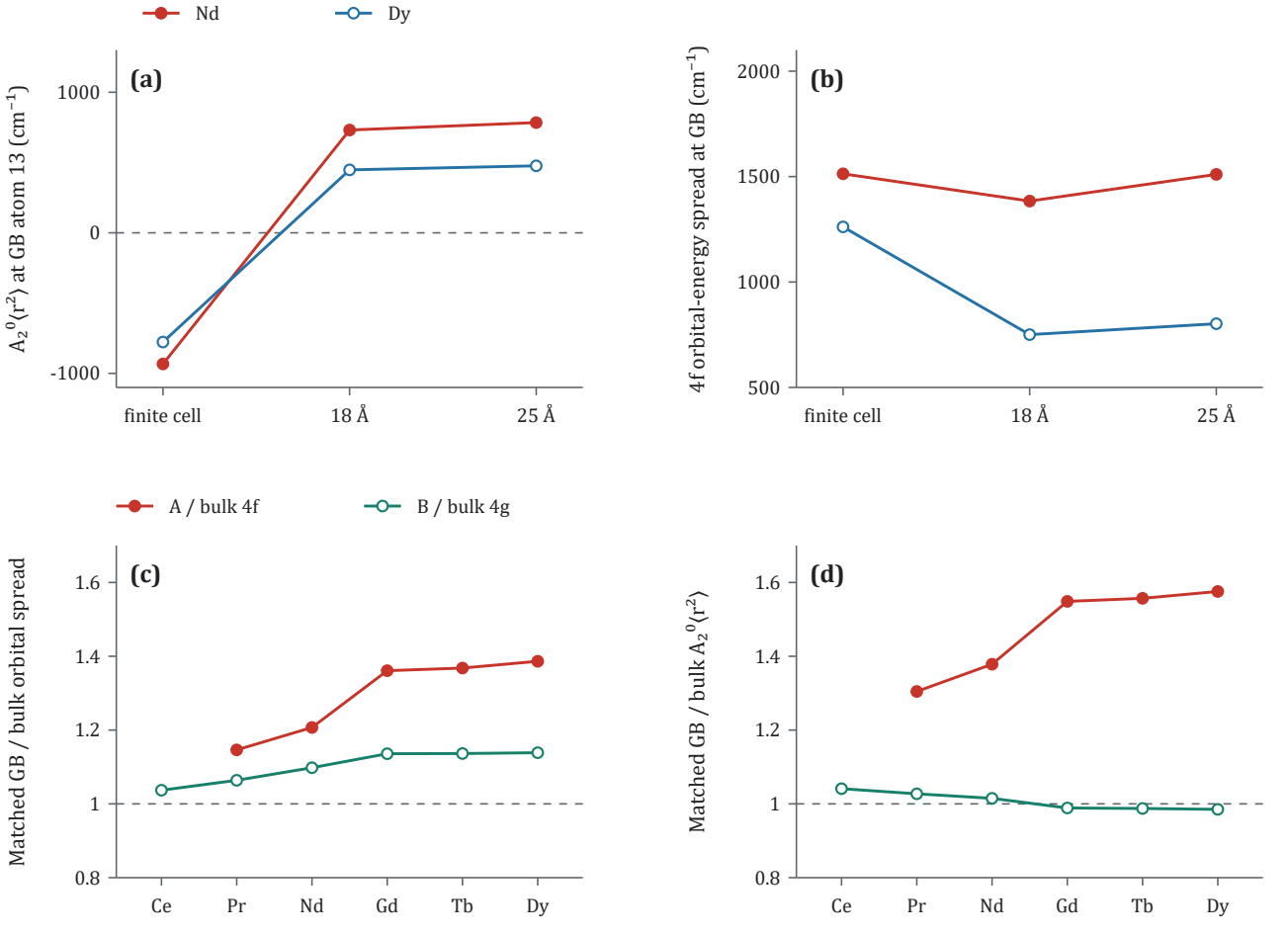

Nd
Dy
(a)
$A_2^0\langle r^2\rangle$ at GB atom 13 (cm^{-1})
1000
0
-1000
finite cell
18 Å
25 Å
(b)
4f orbital-energy spread at GB (cm^{-1})
2000
1500
1000
500
finite cell
18 Å
25 Å
A / bulk 4f
B / bulk 4g
(c)
Matched GB / bulk orbital spread
1.6
1.4
1.2
1
0.8
Ce
Pr
Nd
Gd
Tb
Dy
(d)
Matched GB / bulk $A_2^0\langle r^2\rangle$
1.6
1.4
1.2
1
0.8
Ce
Pr
Nd
Gd
Tb
Dy